

Beyond Slices: A Narrative-Driven, Multi-User, Multi-Modal Virtual Reality System for Medical Imaging Presentation

Sinha Afroz^{1*} Devon Haynes^{1†} Reece Freeman^{1‡} Mason Prather^{1§} Feiyu Lu^{2¶}
 Brooke Zhao^{1||} Joy Li^{1**} Sungchul Jung^{1††} Lei Zhang^{1‡‡}

¹Kennesaw State University ²Independent Researcher



Figure 1: 3D UI techniques designed by the project.

ABSTRACT

Patient education in pediatric oncology is particularly challenging when radiologists and surgeons rely on conventional 2D diagnostic imaging to explain tumors to patients and their families. Standard MRI and CT scans present anatomy as grayscale image slices, making it difficult for non-experts to understand the tumor's three-dimensional structure and its spatial relationship to surrounding tissues. In addition, caregivers frequently ask “why and how did this happen?”, requiring explanations that go beyond visualizing anatomical abnormalities. This work proposes a narrative-driven, multi-user extended reality (XR) approach that presents 3D anatomical reconstructions derived from 2D imaging data. The system emphasizes a shared-viewpoint design to support collaborative understanding and introduces a set of custom 3D user interface (UI) techniques, including opacity adjustment, linked 2D–3D representations, zoom-based animated disease progression, and haptic-enhanced virtual touch, to help clinicians contextualize anatomical and causal relationships when addressing caregivers’ questions.

Index Terms: Extended Reality (XR), Virtual Reality (VR), 3D User Interfaces (3DUI), Medical Visualization, Tumor Visualization

1 INTRODUCTION

Effective patient education is a critical component of pediatric oncology care, yet it remains particularly challenging when clinicians must communicate complex diagnostic imaging information to young patients and their families. Radiologists and surgeons

commonly rely on conventional two-dimensional (2D) diagnostic modalities such as magnetic resonance imaging (MRI) and computed tomography (CT) scans during clinical consultations. These scans present anatomy as grayscale cross-sectional slices, which are difficult for non-expert audiences to interpret. While prior research has demonstrated that reconstructing 3D anatomy from 2D data enhances clinical understanding and surgical planning [1], most applications remain confined to screen-based visualization. Although useful for trained professionals, these systems often fail to fully leverage embodied spatial cues, depth perception, scale, and immersion, which are essential for an intuitive understanding of complex anatomical structures.

In recent years, extended reality (XR) technologies, particularly virtual reality (VR), have emerged as promising tools for enhancing cancer patient education. Several studies in radiation oncology have demonstrated that VR-based visualization of patient-specific imaging data can improve patient understanding, satisfaction, and engagement compared to traditional consultation methods [2, 3].

Our design solutions were informed by conversations with a radiologist who pioneers the use of 3D models generated from 2D diagnostic images in pediatric patient education. Our interviews with him identified two major design goals (DG) for multi-user immersive presentation systems the hospital uses for patient education:

DG1. Extended 3D UI Support: As a multi-user VR system, it lacks a shared viewpoint among users, which hinders collaborative understanding when a surgeon interacts with a 3D model during a presentation. In addition, when users examine anatomical structures, the system provides no mechanism for resolving occlusion between overlapping organs.

DG2. Narrative-Driven Medical Information Presentation: Beyond spatial comprehension, pediatric oncology consultations frequently involve emotionally and cognitively demanding questions from caregivers, most notably, “Why and how did this happen?” Addressing these requires more than identifying abnormalities; it necessitates contextual explanations of tumor location, growth patterns, and interactions with surrounding tissues. However, conventional imaging presentations provide limited support for conveying these anatomical and causal relationships in an accessible manner.

*email: safroz1@students.kennesaw.edu

†email: dwayne33@students.kennesaw.edu

‡email: rfreem57@students.kennesaw.edu

§email: mprath10@students.kennesaw.edu

¶email: feiyulu@vt.edu

||email: yzhao20@kennesaw.edu

**email: yli49@kennesaw.edu

††email: sjung11@kennesaw.edu

‡‡email: lzhang24@kennesaw.edu

2 DESIGN CONCEPT

Our design concept focused on two categories of improvements to extend the current system's functionality for immersive, narrative-driven diagnostic imaging presentations. The first category involves extended 3D UI designs to enhance information presentation in VR. These include shared viewpoint support for a multi-user VR system, adjustable opacity for anatomical structures, and visual connectors that link anatomical objects in 2D CT scans to their corresponding structures in the 3D model. The second category introduces additional 3D UI features to support a narrative-based review of diagnostic information (e.g., tumor growth patterns and invasiveness) with patients and their families. These features include animated tumor growth visualizations through a zoom-in interaction and haptic feedback that simulates virtual touch of anatomical structures for patient family members. The implementation of these design concepts is described in the following sections.

3 SYSTEM AND 3D USER-INTERFACE DESIGN

We implemented our prototype using a Unity multi-user VR project built with Netcode for GameObjects. CT scan images and the corresponding 3D anatomical models were obtained from a sample dataset available through the 3D Slicer Imaging Computing Platform. The finalized system was deployed on the Meta Quest 3. In the multi-user setup, one user assumes the role of the surgeon (U1), while the other takes the role of a patient family member (U2). At the start of the experience, each user receives role-specific task instructions. When the experience starts, U1 follows audio prompts to present diagnostic information to U2. During the presentation, U2 is prompted to ask questions at key moments. These questions trigger interactive tasks that advance the narrative flow of the experience and guide the progression of the diagnostic explanation.

3.1 DG1. Extended 3D UI Support

3.1.1 Shared Viewpoint for Presentation.

A core challenge in collaborative VR environments is maintaining a shared focus and perspective when users move independently in both physical and virtual spaces. Prior work on shared-viewpoint 3DUI designs suggests synchronizing the presenter's camera or viewpoint with that of the audience using specific interface techniques [4]. Inspired by real-world surgical practices, where surgeons' viewpoints often broadcast to large surgical displays to support collaboration and education, we implemented a large virtual screen within the virtual environment (VE). This screen displays the surgeon's viewpoint during diagnostic explanations, allowing all participants in the virtual lobby to share the surgeon's perspective while they present and interact with the 3D anatomical model.

3.1.2 Opacity Adjustment for Anatomical Objects.

Occlusion is a common issue when viewing 3D anatomical models generated from CT scans due to overlapping organs and tissues. To address this, we implemented a 3D UI technique that allows users to adjust the opacity of individual anatomical structures upon selection. This approach enables users to reveal internal structures without separating the entire model, thereby preserving spatial context and anatomical relationships. In implementation, when an anatomical object is hovered over, it is highlighted with a glow. The user may then adjust the object's opacity with a controller joystick. Hovering over a tumor occludes the surrounding anatomical structures to emphasize the tumor and improve focus.

3.1.3 Linking CT Scan Images and 3D Anatomical Model.

The hospital's existing VR visualization system does not support the simultaneous display of CT scans and 3D anatomical models within the same virtual environment. Consequently, users cannot directly associate specific anatomical features visible in 2D CT

slices with their corresponding structures in the 3D model. To address this limitation, we introduced interface elements that visually link highlighted regions in CT scans with the corresponding anatomical objects in the 3D model. This integration supports a more coherent understanding of diagnostic information by bridging 2D and 3D representations.

3.2 DG2. Narrative-Driven Diagnostic Info Presentation

To address the contextual causality challenges in pediatric oncology consultations, we extended the diagnostic imaging presentation with additional narrative-supportive features.

3.2.1 Two-Handed Zoom-In Tumor Growth Visualization.

We designed a two-handed "stretch-to-scale" zoom-in 3D UI that enables users to scale up one spot in the anatomical model with two controllers and view an animated simulation of tumor cell growth and proliferation within surrounding tissues. We hypothesize that this visualization can convey tumor characteristics such as growth patterns and invasiveness, thereby supporting surgeons' explanations of disease progression and surgical decision-making. In addition, before-and-after visualizations of surgical regions are presented to illustrate how surgical intervention may alter anatomical structures and contribute to treatment outcomes.

3.2.2 Haptic-Enhanced Virtual Touch Interaction.

We implemented a dedicated haptic interaction design that integrates controller vibration, proximity-based visual feedback, and localized deformation. When users touch an anatomical object using a virtual hand, tactile sensation of force is simulated through controller vibration, while a deformation effect is rendered on the geometry using a custom shader to enhance the perception of force touch. This approach provides real-time, localized feedback that visually reinforces touch during XR interaction. The inclusion of this interaction was informed by interviews with the radiologist, who noted that some families requested copies of their child's diagnostic 3D models. These models were often treated as digital legacies, particularly in cases of terminal diagnoses or poor prognoses. The haptic-enhanced interaction aims to support a embodied and emotionally meaningful engagement with these digital representations.

4 CONCLUSION

In this project, we presented a novel multi-user VR-based system design that extends existing 3D diagnostic imaging visualization tools through a set of integrated 3D UI techniques. Our design addresses limitations in shared perspective, spatial comprehension, and narrative explanation commonly encountered in pediatric oncology consultations. We believe this work contributes to the emerging research on designing core system features for multi-user XR-based medical imaging visualization in oncology, with the goal of improving both informational clarity and user experience for patients, families, and clinicians.

REFERENCES

- [1] G. Pelizzo, U. M. Pierucci, M. Marinaro *et al.*, "Virtual reality for pre-operative planning and education in pediatric surgery: Preliminary results for the treatment of congenital malformations and tumors," *World Journal of Surgery*, vol. 49, no. 6, pp. 1497–1507, 2025. 1
- [2] L. J. Wang, B. Casto, J. Y. Luh *et al.*, "Virtual reality-based education for patients undergoing radiation therapy," *Journal of Cancer Education*, vol. 37, no. 3, pp. 694–700, 2022. 1
- [3] D. E. Holt, A. L. Carr, S. Roberts *et al.*, "The use of virtual reality for 3d diagnostic imaging review enhances understanding and education of patients with cancer," *International Journal of Radiation Oncology Biology Physics*, vol. 122, no. 5, pp. 1102–1112, 2025. 1
- [4] S. Valin, A. Francu, H. Trefftz, and I. Marsic, "Sharing viewpoints in collaborative virtual environments," in *Proceedings of the 34th Annual Hawaii International Conference on System Sciences*. IEEE, 2001. 2